

Inducing and Quantifying *Clostridium difficile* Spore Formation.

PMID: 27507338 | Indexed in PubMed

The Gram-positive nosocomial pathogen *Clostridium difficile* induces sporulation during growth in the gastrointestinal tract. Sporulation is necessary for this obligate anaerobe to form metabolically dormant spores that can resist antibiotic treatment, survive exit from the mammalian host, and transmit *C. difficile* infections. In this chapter, we describe a method for inducing *C. difficile* sporulation in vitro. This method can be used to study sporulation and maximize spore purification yields for a number of *C. difficile* strain backgrounds. We also describe procedures for visualizing spore formation using phase-contrast microscopy and for quantifying the efficiency of sporulation using heat resistance as a measure of functional spore formation.

Assembly of the exosporium layer in *Clostridioides difficile* spores.

PMID: 35182899 | Indexed in PubMed

Clostridioides difficile is a Gram-positive, spore-forming obligate anaerobe and a major threat to the healthcare system world-wide. Because of its strict anaerobic requirements, the infectious and transmissible morphotype is the dormant spore. During infection, *C. difficile* produces spores that can persist in the host and are responsible for disease recurrence and transmission, especially between hospitalized patients. Although the *C. difficile* spore surface mediates critical interactions with host surfaces, this outermost layer, known as the exosporium, is poorly conserved when compared to members of the *Bacillus* genus. Notably, the exosporium has been shown to be important for the persistence of *C. difficile* in the host. In this review, the ultrastructural properties, composition, and morphogenesis of the exosporium will be discussed.

Clostridioides difficile and the Microbiota Early in Life.

PMID: 34791400 | Indexed in PubMed

Clostridioides difficile is a spore-forming, obligate anaerobe, and ubiquitous nosocomial pathogen. While *C. difficile* infection in adults causes a spectrum of disease, including pseudomembranous colitis and toxic megacolon, healthy infants are asymptotically colonized at high rates. The mechanisms leading to high colonization rates and infant protection from *C. difficile* are currently unknown; however, the ecology and metabolic state of the intestinal microbiome are factors known to influence *C. difficile* pathogenesis. In this review, we will examine the aspects of the early-life microbiome that may contribute to the incidence of *C. difficile* and protection from disease manifestation in infants. We will also discuss whether features of the adult microbiota that enable and restrict *C. difficile* are prevalent during early-life colonization.

Comparative Genomics of *Clostridioides difficile*.

PMID: 38175477 | Indexed in PubMed

Clostridioides difficile, a Gram-positive spore-forming anaerobic bacterium, has rapidly emerged as the leading cause of nosocomial diarrhoea in hospitals. The availability of large numbers of genome sequences, mainly due to the use of next-generation sequencing methods, has undoubtedly shown their immense advantages in the determination of *C. difficile* population structure. The implementation of fine-scale comparative genomic approaches has paved the way for global transmission and recurrence studies, as well as more targeted studies, such as the PaLoc or CRISPR/Cas systems. In this chapter, we provide an overview of recent and significant findings on *C. difficile* using comparative genomic studies with implications for epidemiology, infection control and understanding of the evolution of *C. difficile*.

Clostridioides difficile toxins: mechanisms of action and antitoxin therapeutics.

PMID: 34837014 | Indexed in PubMed

Clostridioides difficile is a Gram-positive anaerobe that can cause a spectrum of disorders that range in severity from mild diarrhoea to fulminant colitis and/or death. The bacterium produces up to three toxins, which are considered the major virulence factors in *C. difficile* infection. These toxins promote inflammation, tissue damage and diarrhoea. In this Review, we highlight recent biochemical and structural advances in our understanding of the mechanisms that govern host-toxin interactions. Understanding how *C. difficile* toxins affect the host forms a foundation for developing novel strategies for treatment and prevention of *C. difficile* infection.

Clostridioides difficile infection: microbe-microbe interactions and live biotherapeutics.

PMID: 37228365 | Indexed in PubMed

Clostridioides difficile is a gram-positive, spore-forming, obligate anaerobe that infects the colon. *C. difficile* is estimated to cause nearly half a million cases in the United States annually, with about 29,000 associated deaths. Unfortunately, the current antibiotic treatment is not ideal. While antibiotics can treat the infections, they also disrupt the gut microbiota that mediates colonization resistance against enteric pathogens, including *C. difficile*; disrupted gut microbiota provides a window of opportunity for recurrent infections. Therefore, therapeutics that restore the gut microbiota and suppress *C. difficile* are being evaluated for safety and efficacy. This review will start with mechanisms by which gut bacteria affect *C. difficile* pathogenesis, followed by a discussion on biotherapeutics for recurrent *C. difficile* infections.

Clostridium difficile infection: a worldwide disease.

PMID: 24516694 | Indexed in PubMed

Clostridium difficile, an anaerobic toxigenic bacterium, causes a severe infectious colitis that leads to significant morbidity and mortality worldwide. Both enhanced bacterial toxins and diminished host immune response contribute to symptomatic disease. *C. difficile* has been a well-established pathogen in North America and Europe for decades, but is just emerging in Asia. This article reviews the epidemiology, microbiology, pathophysiology, and clinical management of *C. difficile*. Prompt recognition of *C. difficile* is necessary to implement appropriate infection control practices.

Clostridioides difficile epidemiology in India.

PMID: 35063600 | Indexed in PubMed

Clostridioides difficile infection (CDI) continues to affect hospitalized patients and community populations worldwide. In contrast to the substantial resources invested in the diagnosis and prevention of CDI in high-income countries, this anaerobic toxigenic bacterium has been largely overlooked in low-and-middle-income countries (LMICs) such as India, where there remains a paucity of epidemiologic data evaluating the burden of CDI. Extensive multi-institutional studies describing *C. difficile* epidemiology in India have not yet been performed. Given recent economic growth in many Asian countries, with aging populations, increased access to healthcare and widespread inappropriate use of antimicrobials, *C. difficile* is likely to be highly prevalent and causing significant disease burden. Greater efforts are required to enhance awareness of this neglected pathogen, through educating healthcare practitioners to test for CDI. There is also an urgent need to strengthen laboratory capacity, and ideally establish a national reference laboratory, to help facilitate a greater understanding of the molecular epidemiology of CDI in India and other LMICs. This mini-review aims to summarize the existing research evaluating the burden of CDI in humans and the environment in India.

Burden of Clostridioides difficile infection (CDI) - a systematic review of the epidemiology of primary and recurrent CDI.

PMID: 34016040 | Indexed in PubMed

Clostridioides difficile is a Gram-positive anaerobic bacterium, which causes *Clostridioides difficile* infection (CDI). It has been recognised as a leading cause of healthcare-associated infections and a considerable threat to public health globally. This systematic literature review (SLR) summarises the current evidence on the epidemiology and clinical burden of CDI. A SLR was conducted to identify CDI and recurrent CDI (rCDI) epidemiology studies, to evaluate patient and disease characteristics, incidence rates, epidemiological findings and risk factors. Embase, MEDLINE and the Cochrane Library databases were searched for English articles from 2009 to 2019. Included territories were the United Kingdom, France, Germany, Italy, Spain, Poland, US, Canada, Australia, Japan and China. Of 11,243 studies identified, 165 fulfilled the selection criteria. An additional 20 studies were identified through targeted review of grey literature. The most widely reported findings were incidence and risk factors for CDI and rCDI. Among key studies reporting both healthcare-associated (HA-CDI) and community-associated CDI (CA-CDI) incidence rates for each country of interest, incidence rates per 10,000 patient days in the US were 8.00 and 2.00 for HA-CDI and CA-CDI, respectively. The highest incidence in Europe was reported in Poland (HA-CDI: 6.18 per 10,000 patient days, CA-CDI: 1.4 per 10,000 patient days), the lowest from the UK, at 1.99 per 10,000 patient days and 0.56 per 10,000 patient days for HA-CDI and CA-CDI, respectively. No clear trend for incidence over time emerged, with most countries reporting stable rates but some either a decrease or increase. Rates of recurrent CDI varied based on geographical setting. The rate of recurrence was lower in community-associated disease compared to healthcare-associated disease. Independent CDI risk factors identified common to both initial CDI and recurrent CDI included increasing age, antibiotic use, recent hospitalisation, and proton pump inhibitor (PPI) use. In addition, leukocyte count, length of hospital stays, and Charlson comorbidity index score featured as statistically significant risk factors for recurrent CDI, but these are not reported among the most common statistically significant risk factors for initial CDI. Despite considerable heterogeneity, evidence suggests substantial

incidence of recurrent and primary CDI, even after considerable efforts in the last decade.

Convenient Protocol for Production and Purification of *Clostridioides difficile* Spores for Germination Studies.

PMID: 33111107 | Indexed in PubMed

Clostridioides difficile, an obligate anaerobic bacterium, causes infections leading to prolonged diarrhea. The bacterium produces dormant spores that can withstand an aerobic environment, resulting in easy environmental transfer. Here, we present a convenient sporulation and purification protocol that can be practiced in any lab setting using a portable anaerobic glove bag. This protocol also optimizes existing cell growth methods and presents a detailed trouble shooting guide. This protocol is a modification of those previously reported by Edwards and McBride (2016) and Shen et al. (2016).
